Lecture 2 - ODEs in the S-Domain & Feedback

Instructor: Martin Nisser
Scribe: Catherine Ingram (Please send feedback to ingramca@uw.edu)

2.1 Solving ODE Systems In The Frequency Domain

We have some system, e.g. a spring-mass-damper, that we consider linear, time-invariant (LTI) shown in Figure 2.1.

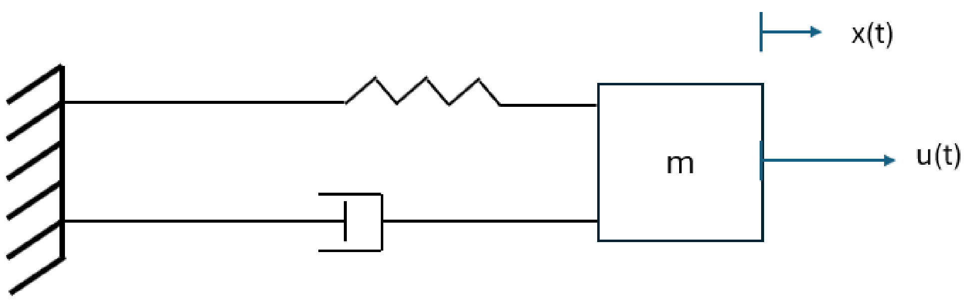
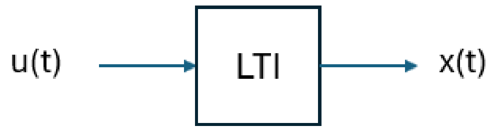


Figure 2.1: Spring-mass-damper system with position of the mass x(t) and control input u(t).

The equations of motion for this system, using Newtons Second Law,

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = u(t), \tag{2.1}$$

where k is the spring stiffness, b is the damper coefficient, m is the mass, and u is the input. We want to find the solution x(t) given some input u(t) to an LTI system. The general problem setup for an ODE problem in time is demonstrated in Figure 2.2.



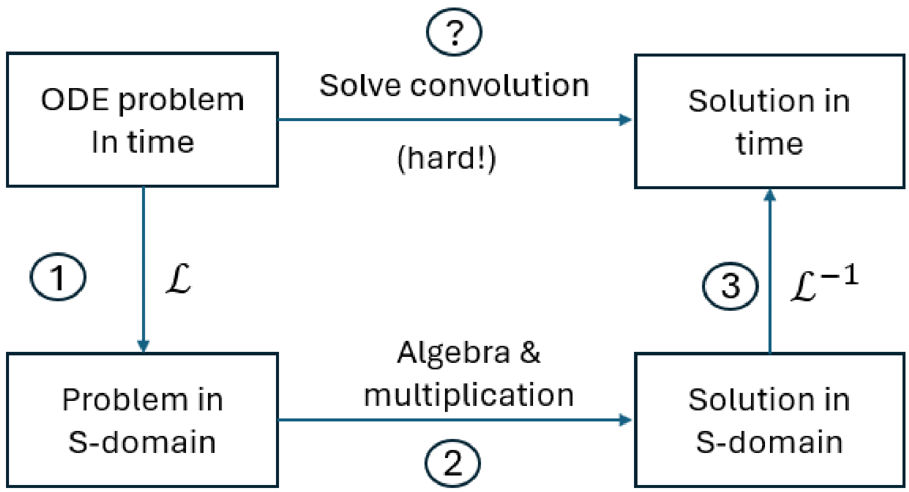


Figure 2.2: Problem setup.

But, solving this directly involves a difficult step of solving the convolution integral,

$$\boxed{x(t) = \int_0^t u(\tau)\tilde{x}(t-\tau)\,d\tau} \tag{2.2}$$

In the convolution integral, $u(\tau)$ is the input and $\tilde{x}(t-\tau)$ is the impulse response.

Instead: Take the Laplace transform, do algebra, and take the inverse Laplace to return to the time domain.

1: Take the Laplace Transform

- We use a change of variable from t to s using the Laplace transform

$$\rightarrow \quad \mathcal{L}\{f(t)\}(s) = \int_0^\infty f(t)e^{-st}\,dt \quad \rightarrow \quad \text{But we don't have to solve this}$$

$$\rightarrow \quad \text{Laplace Tables, e.g.} \quad u(t) = e^{-at} \quad \rightarrow \quad U(s) = \frac{1}{s+a}$$

- We can also use Laplace tables for derivatives of signals (our equations of motion):

$$\rightarrow \quad \mathcal{L}\{\frac{df}{dt}\}(s) = \int_0^\infty (\frac{df}{dt})e^{-st}\,dt$$

$$\rightarrow \quad \underline{\text{Laplace Tables}}: \ \frac{dx}{dt} \xrightarrow{\mathcal{L}(\cdot)} sX(s) - x(0), \quad x(0) \text{ is the initial condition}$$

- **Tables for common transforms are uploaded on canvas**.

2: Solve for X(s) using algebra

- Convolution in the time domain becomes multiplication in the s-domain.

$$X(s) = \mathcal{L}\{(*)\} = \mathcal{L}\{(u * \tilde{x})(t)\} = U(s)\tilde{X}(s)$$

For clarity :

$$X(s) = U(s)\tilde{X}(s), \quad \text{where } \tilde{X}(s) \text{ is the Laplace of the impulse response}$$

- We call $\tilde{X}(s)$ the Transfer Function since multiplying it by U(s) gives us X(s).
- Traditionally, this is denoted with the letter G or H, e.g. G(s) = $\tilde{X}(s)$.
- Transfer Functions are very useful for block diagrams.
  - Instead of: $u(t) \rightarrow \boxed{LTI} \rightarrow x(t)$
  - We now have: $U(s) \rightarrow \boxed{G(s)} \rightarrow X(s)$

    Where $G(s) = \frac{X(s)}{U(s)}$

3: Finally, take the inverse Laplace

- Change of variable from s back to t using $\mathcal{L}^{-1}$

$$\mathcal{L}^{-1}\{F(s)\} = \frac{1}{2ni}\lim_{T\to\infty}\int_{\gamma-iT}^{\gamma+iT} e^{st}F(s)\,ds \quad \rightarrow \quad \text{Again, we } \underline{\text{use Laplace tables}}.$$

$$\rightarrow \text{e.g. :} \quad U(s) = \frac{1}{s+a} \xrightarrow{\mathcal{L}^{-1}(\cdot)} u(t) = e^{-at}$$

Note: We are not concerned with solving these integrals in this course, but it is good to remember where the Laplace table comes from.

Example

Find the solution to the differential euqation:

$$\ddot{y}(t) + 5\dot{y}(t) + 4y(t) = 2e^{-2t}, \qquad \text{where : } u(t) = 2e^{-2t},\ y(0) = 0,\ \dot{y}(0) = 0$$

Solution

1: Take Laplace transform:

$$\text{Using :} \quad \mathcal{L}(\ddot{y}) = s^2Y(s) - sy(0) - \dot{y}(0)$$

$$\mathcal{L}(\dot{y}) = sY(s) - y(0)$$

$$\mathcal{L}(2e^{-2t}) = \frac{2}{s+2}$$

$$\rightarrow s^2Y(s) - s\cdot 0 - 0 + 5[sY(s) - 0] + 4Y(s) = \frac{2}{s+2}$$

2: Solve for Y(s):

$$
\begin{aligned}
Y(s)[s^2+5s+4] &= \frac{2}{s+2} \\
Y(s)(s+1)(s+4) &= \frac{s}{s+2} \\
Y(s) &= \underbrace{\frac{1}{(s+1)(s+4)}}_{G(s)} \cdot \underbrace{\frac{2}{(s+2)}}_{U(s)}
\end{aligned}
\qquad
\left.\begin{aligned}
&Y(s) = G(s)U(s) \\
&U(s) \rightarrow \boxed{G(s)} \rightarrow Y(s)
\end{aligned}\right.
$$

Separate terms by taking Partial Fractions (so we can use Laplace tables for y(t))

$$
\begin{aligned}
&Y(s) = \frac{2}{(s+1)(s+4)(s+2)} = \frac{A}{s+1} + \frac{B}{s+4} + \frac{C}{s+2} \\
&\rightarrow \text{Solve for A, B, C} \\
&\rightarrow Y(s) = \frac{2/3}{s+1} + \frac{1/3}{s+4} - \frac{1}{s+2}
\end{aligned}
$$

3: Take inverse Laplace (from tables)

$$\therefore \quad y(t) = \frac{2}{3}e^{-t} + \frac{1}{3}e^{-4t} - e^{-2t}$$

2.2 Stability in Open and Closed Loop

2.2.1 Motivation: Stability in Open-Loop

- In the (open-loop) example above, we knew that Y(s) was stable for bounded inputs U(s) because the poles of G(s) were in the LHP (poles are the roots of the characteristic equation).
- Let's redraw the block diagram in Figure 2.3 so we can see the reference R:

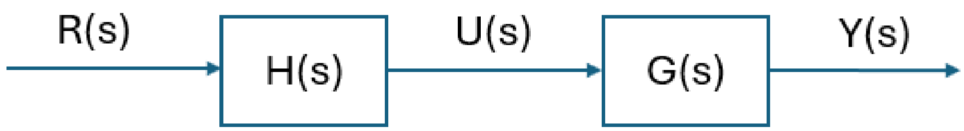


Figure 2.3: Open-loop block diagram.

Where: $Y(s) = G(s)U(s) = G(s)\overbrace{H(s)R(s)}^{U(s)}$

And: $H(s)$ is our controller, which we design

- The open-loop transfer function (from R to Y) is:

$$\frac{Y(s)}{R(s)} = G(s)H(s)$$

→ Again, BIBO stability is ensured if the poles of G(s)H(s) are in the LHP.

- So, if our plant G(s) has a RHP pole, can we design H(s) to make the T.F. stable?
  - Naively: Try and cancel the RHP pole in G using RHP zero in H

$$GH = \underbrace{\frac{1}{(s-1)}}_{G} \cdot \underbrace{\frac{(s-1)}{1}}_{H} \quad \rightarrow \quad \textbf{Bad}$$

  - NO! Because,
    * Any model inaccuracy in G will give instability!
    * A disturbance W yields the term WG, so the RHP survives no matter the choice for H.
    * Also, adding a RHP zero gives bad performance (more on this later).

∴ $\boxed{\text{Open-loop controllers cannot make an unstable plant be stable}}$.

2.2.2 Stability for Closed-Loop

- Lets add feedback and redraw our closed-loop system block diagram in Figure 2.4:

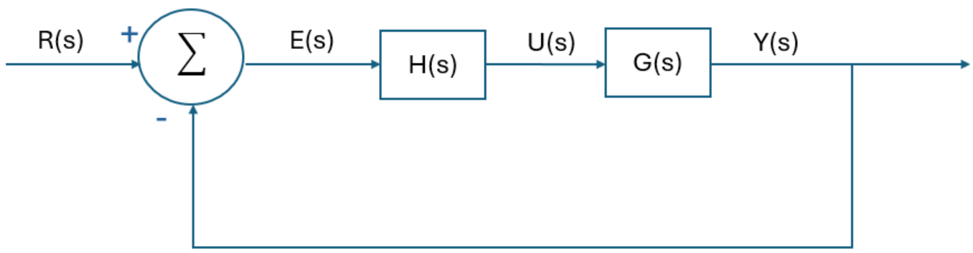


Figure 2.4: Closed-loop block diagram.

- Find the new transfer function $R \to Y$: (dropping the "(s)" for clarity)

$$\begin{aligned}
&Y = GHE = GH(R - Y) = GHR - GHY \\
&Y + GHY = GHR \\
&Y(1 + GH) = GHR \\
&Y = \frac{GH}{1+GH}R \quad \to \quad \frac{Y}{R} = \frac{GH}{1+GH}
\end{aligned}$$

- For stability, we require the poles of 1 + GH to be in the LHP:
  - Define polynomials a(s), b(s), c(s), d(s) such that $G = \frac{b}{a}$, $H = \frac{d}{c}$.
  - The characteristic equation is

$$\begin{aligned}
&1 + GH = 0 \\
&1 + \frac{bd}{ac} = 0 \\
&ac + bd = 0
\end{aligned}$$

$\therefore$ We still can't use a RHP zero in H to cancel a RHP pole in G, however, the separation of the terms means that a RHP pole in G does not prevent the design of a controller that makes the system stable.

2.3 Regulation: Disturbance and Noise Rejection

2.3.1 Recap of the 4 Control Objectives

We've just seen that closed-loop feedback has advantages over open-loop control in terms of stability. In general, there are 4 things we care about:

- Stability (stable output, discussed above)
- Regulation (reject disturbances)
- Sensitivity (Robustness to plant changes)
- Tracking ($E_{ss} = 0$)

2.3.2 Regulation

Let's add a disturbance (W) and noise (V) to our closed-loop block diagram and see how we might limit their effect on the output in Figure 2.5.

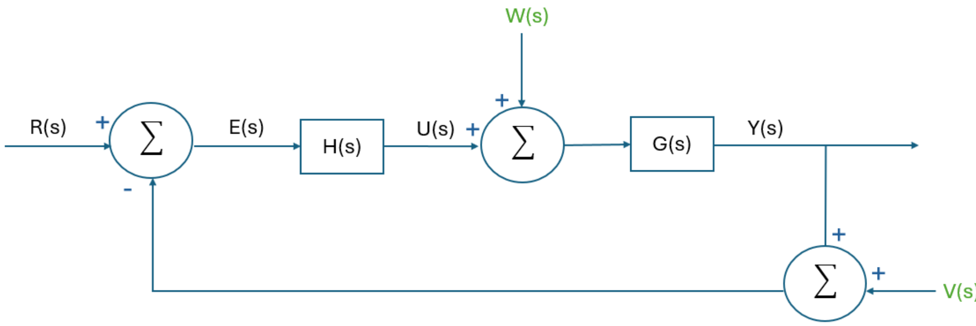


Figure 2.5: Closed-loop block diagram with disturbance W and noise V.

There are now 3 external inputs to our system:

- The Reference R, which we want the output to track.
- Disturbance W, which our controller must counteract.
- Noise V, which our controller should ignore.

Earlier, we derived the transfer function $R \to Y$ to be $\frac{GH}{1+GH}$.

$\to$ If we do the same for W $\to$ Y and $V \to Y$, the equation for the output is just the superposition of its response to the individual inputs:

$$Y = \frac{GH}{1+GH}R + \frac{G}{1+GH}W - \frac{GH}{1+GH}V$$

$$\text{where we define} \quad T = \frac{GH}{1+GH} \quad \} \text{ Transfer Function}$$

$$S = \frac{1}{1+GH} \quad \} \text{ Sensitivity Function}$$

$$\to T + S = 1$$

$$\to Y = TR + GSW - TV$$

Note that there appears to be a trade-off between rejecting W and V:

- If $|H|$ is large $\to$ W is attenuated ✓
  $\to$ V is unity ×
- If $|H|$ is small $\to$ W is amplified ×
  $\to$ V is attenuated ✓

Solution is based on the fact that W & V are frequency dependent.

- The disturbances W are typically low frequency, even DC.

  - For launch vehicles: wind gusts, gravity, structural modes (1st bending and torsion), fuel slosh modes, etc.
  - Satellites: gravity perturbations (J2 effect), solar panel modes, propellant slosh modes, solar radiation pressure, etc.
  - Landers: gravity, unmodeled aerodynamics (Mars density gradient), landing legs modes, etc.
- The noise V is typically high frequency, and should have no bias by construction. These typically come from your sensors, which sample at high frequencies and often have high frequency noise. This can also come from higher frequency vibrations on the vehicle, which you typically just want to ignore as they are unavoidable.

We just discussed the two of the four objectives of control (**stability** and **regulation**). Next lecture, we'll discuss the last two (**sensitivity** and **tracking**).